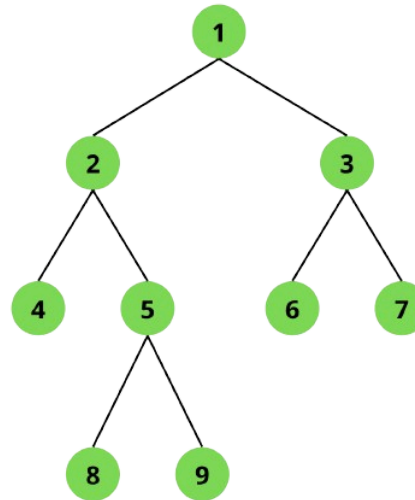


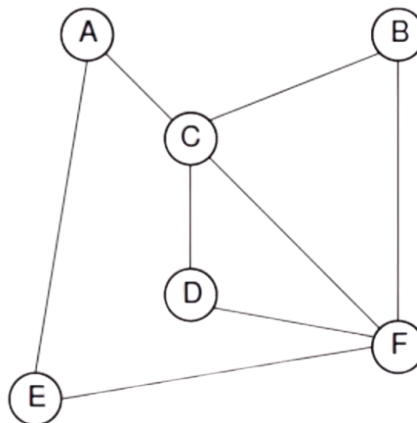
concept of AND/OR graphs in problem-solving and game tree search. Differentiate between AND nodes and OR nodes with clear examples. (T)

Three types of Depth First Search (DFS) traversals on binary trees: preorder, inorder, and

postorder. Given a binary tree (P)



Breadth First Search (BFS) algorithm on a graph (Source – A). (P)



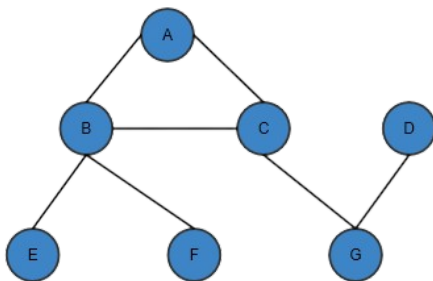
Fundamental concepts of P and NP complexity classes. Explain with examples what it means for a problem to be in P (polynomial time solvable) versus NP (polynomial time verifiable). (T)

Vertex Cover problem as an NP-complete problem. Examine the proof structure: analyze why Vertex Cover belongs to NP by explaining the verification process.(T)

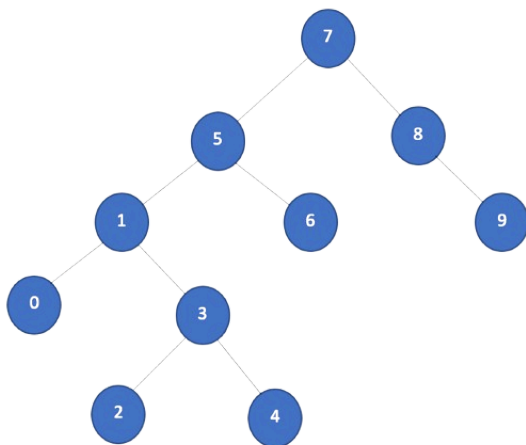
The differences between P, NP, NP-Hard, and NP-Complete complexity classes (T)

Concept of biconnected components in an undirected graph. Differentiate between articulation points (cut vertices) and biconnected components. (T)

Illustrate the Breadth First Search (BFS) algorithm on a graph (Source – A). (P)



Apply Depth First Search (DFS) traversals: preorder, inorder, and postorder (P)



Illustrate the concept of polynomial-time reduction with a practical example. Show how one problem can be transformed into another problem in polynomial time.(T)

Examine the relationship between three classical NP-complete graph problems: Clique, Independent Set, and Vertex Cover. (T)

Classify the Hamiltonian Cycle problem and Traveling Salesman Problem (TSP) within the complexity hierarchy. (T)

Explain the concepts of NP, NP-Hard, and NP-Complete classes. Illustrate the relationships among these classes with a diagram and provide real-world examples of each class. (T)

Discuss the significance of polynomial-time reductions in the theory of NP-Completeness. How does reduction help prove that a problem is NP-Complete or NP-Hard? Give an example to support your answer (T)

What are the necessary conditions for a problem to be classified as NP-Complete? Use graph-based problems such as the Clique or Hamiltonian Cycle problem as examples to illustrate these conditions (T)

Differentiate between decision, optimization, and search problems in the context of NP-Hard and NP-Complete classes. Describe with reference to graph problems like Vertex Cover and Traveling Salesman Problem. (T)

Describe the concept of polynomial-time verifiability and its importance in defining NP problems. Use examples from graph theory, such as 3-Coloring or Independent Set, to support your explanation. (T)

Explain why finding efficient (polynomial-time) algorithms for NP-Complete problems is a major challenge in computer science. What are the practical implications if a polynomial-time solution for an NP-Complete problem is found? (T)